

Question				Marking details	Marks available				Maths	Prac
					AO1	AO2	AO3	Total		
	(c)	(i)		U value = Energy per second / (area $\times$ temperature difference) / rate of heat transfer per unit area per unit temperature difference / $U = \frac{P}{A\Delta\theta}$ with symbols correctly identified (1) Thickness of material or conductivity of material (1)	2			2		
		(ii)		Energy per second lost = $1.8 \times (5 \times 3) \times (21 - 9)$ (1) for substitution Energy per second = 324 [W] (1)		2		2	2	
		(iii)		$1\,000 = 1.8 \times 15 \times (21 - T)$ (1) $T = -16$ [°C] (1)		2		2	2	
				Question 9 total	6	9	5	20	10	0